SMART INDIA HACKATHON 2025

Problem Statement ID : 25050
Problem Statement Title : SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION
Theme : TRANSPORTATION & LOGISTICS
PS Category : Software
Team ID : 92770
Team Name : 404 The Optimists

Lanezy: AI-Powered Adaptive Traffic Control - Revolutionizing Indias Roads

Lanezy: AI-Powered Adaptive Traffic Control

Real-world issue: Urban congestion, accidents, emergency delays, poor enforcement cost billions, massive fuel & time loss.
Why important: India loses Rs 60,000+ crores annually; 150,000+ lives lost yearly.
Solution: AI optimizes signals, detects anomalies, ensures safer faster flow.

Core Innovation: ChatBot
Primary: AI-Powered Smart Signal Control | Priority Vehicle Routing
Safety: Wrong Side Driving Fine | Illegal Parking | Accident Button
Monitoring: Pothole | Helmet | Ambulance Override (IoT) | Manual Override | Animal Detection

TECHNICAL APPROACH

METHODOLOGY: Per-Signal Login -> Emergency Override -> Manual Override -> Adaptive Control -> Parallel Alerts
Flow: Camera -> YOLO -> Backend/cloud -> IoT ESP8266 -> Signal
Second: IOT + Lat/Long + Cloud (Haversine) -> Backend -> Signal green for ambulance.

SYSTEM IOT: ESP8266 photos

TECH USED: FrontEnd React+Tailwind | BackEnd FastAPI | Deploy Render+Vercel | Hardware LEDs, GPS, ESP8266, YOLOv8.pt

FEASIBILITY AND VIABILITY

FEASIBILITY: Infrastructure Reusability, Scalable, Cost-Effective, Authority Integration
VIABILITY: Proven Concept, Builds Public Trust, Guarantees Accountability, Ensures Authority Engagement
BUSINESS: Government Cost Savings, New Revenue via e-challan

IMPACT AND BENEFITS

Reduced Congestion 30-40% | Faster Response for ambulance | Lower Fuel Wastage | Emergency Alerts
Safety: quicker response | Economic: time & fuel savings | Environmental: reduced carbon | Technology: AI-driven

Chart: Time Lost 40%->20%, Fuel 25%->10%, Economic 20%->8%, Environmental 10%->4%, Emergency 5%->1%
From Gridlock to Green Light

RESEARCH AND REFERENCES

1. YOLO Vehicle Detection [LINK]
2. Omnidim Notification [LINK]
3. NITI Aayog Smart Cities [LINK]
4. MoRTH Accidents Report [LINK]
5. IEEE Traffic Signal [LINK]
6. EVP [LINK]
7. Adaptive Traffic Control [LINK]

UI/UX: Landing + Dashboard N/S/E/W live YOLO + Emergency SOS 100/108/101